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Questioning the rules of engagement: a critical analysis of the use of limb symmetry index for safe return to sport after anterior cruciate ligament reconstruction

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ABSTRACT

Objective To evaluate the association between limb symmetry index (LSI) in quadriceps and hamstrings strength together with hop tests, as a proxy of recovery, and the deviation from being symmetrical (LSI 100%), with a safe return to sport (RTS) after anterior cruciate ligament reconstruction (ACL-R).

Methods Athletes between 15 and 30 years old with a preinjury Tegner activity level ≥6 were eligible for inclusion. Data were extracted from a rehabilitation-specific registry, Project ACL (Gothenburg, Sweden) at the time of or after RTS for each athlete. The outcome of interest in this study was a safe RTS—not suffering a second ACL injury within 2 years from RTS—addressed by using the LSI in five different ways with results from the test battery which include five tests of muscle function at or after the time of RTS. Logistic regression analyses were performed with safe RTS as a dependent variable and presented with ORs and 95% CIs.

Results In total, 233 athletes (51.1% women) were included. The best-differentiating cut-offs for strength and hop tests showed non-significantly poor discriminatory ability between athletes who had a safe RTS and those who did not (Youden J 0.09–0.24 and area under the curve 0.50–0.59). Athletes who had ≥80% or ≥85% LSI had significantly lower odds of safe RTS compared with athletes who did not meet the cut-offs of safe RTS (OR=0.32 (95% CI 0.12 to 0.87) and OR=0.39 (95% CI 0.18 to 0.84), respectively). There was no effect of a 1% increase in LSI or deviation from 100% symmetry on safe RTS.

Conclusion The use of LSI from tests of muscle function to determine safe RTS after ACL-R, that is, RTS without sustaining a second ACL injury within 2 years, cannot differentiate between athletes who had a safe RTS and those who did not, regardless of whether LSI was used as cut-offs, incremental or as deviation from symmetry. Thus, it is of clinical importance that clinicians do not solely rely on the LSI to clear athletes for RTS.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Various limb symmetry index (LSI) cut-off values for muscle function tests are used to assist in the decision-making process to determine how prepared an athlete is to return to sport (RTS), where the most commonly used cut-off is ≥90% LSI. Although the use of LSI cut-off values has previously not been validated for second anterior cruciate ligament (ACL) injury occurrence, LSI has been accepted based on consensus criteria and expert opinions as an indicator of a successful outcome after ACL reconstruction (ACL-R).

WHAT THIS STUDY ADDS

⇒ This study questions the usefulness of LSI cut-off values to differentiate between athletes with a safe RTS, defined as no second ACL injury, from athletes who sustain a second ACL injury, and discusses the current and future challenges of RTS testing.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Results from this study show that the use of LSI to interpret results from muscle function tests cannot determine a safe RTS, thereby highlighting the necessity to understand the multifaceted nature of RTS following ACL-R. The use of LSI should be valued as one of several components prior to RTS. The results of our study might encourage clinicians to include more parameters than merely LSI when clearing athletes for RTS.

INTRODUCTION

The risk for second anterior cruciate ligament (ACL) injury is notably high when athletes return to sport (RTS) following ACL reconstruction (ACL-R). This risk for second ACL injury is particularly high in individuals <25 years old, with rates ranging from 23% to 35%.[1–3] In an attempt to reduce the proportion of athletes who suffer a second ACL injury and to ensure athletes are adequately prepared for RTS, clinicians are recommended to use muscle strength and hop performance tests, together referred to as muscle function tests and incorporate valid patient-reported outcome measures (PROMs).[4–6] The muscle function tests have been used to assist clinicians in determination if athletes are ready for RTS.[7] Results from muscle function tests are commonly presented as limb symmetry index (LSI), that is, the ratio of results between injured and uninjured limbs expressed as

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a percentage of symmetry.[8] The recommended cut-off is LSI ≥90% across a battery of tests before an athlete is determined ready to RTS with minimal risk for a second ACL injury, that is, a safe RTS.[7] Patients treated for an ACL-R can struggle to pass the RTS criteria in a battery of muscle function tests prior to RTS,[8 9] where passing rates can be as low as 23% between 6 and 36 months follow-up after ACL-R when using LSI ≥90%0.[10] However, Piussi *et al*[11] reported a 50% passing rate in a battery of tests at 12 months after an ACL-R. Grindem *et al*[7] reported that 33% of patients who did not pass ≥90% LSI in quadriceps strength at time for RTS (7.2±2.7 months) suffered a second ACL injury, compared with 13% ACL injuries in patients passing ≥90% in quadriceps strength LSI (10.7±4.3 months). A systematic review from 2019[10] pointed out that passing LSI ≥90% in a battery of tests does not have the ability to predict a second ACL injury whereas most studies reported RTS at 6 months (6–36 months).

Although the use of LSI has been questioned, achieving ≥90% LSI across a battery of tests has previously been associated with a safe RTS.[12] Other studies have not been able to confirm the risk reduction for a second ACL injury.[13 14] To date, it is unclear if LSI cut-offs can differentiate between athletes who have a safe RTS and those who do not. Nevertheless, LSI cut-off values are used in clinical practice to guide physical therapists and athletes in determining and assessing the recovery of symmetrical muscle function and successful outcomes for RTS.[15] The LSI cut-offs have been accepted through consensus statements[16–18] and expert opinions[4 19] but have not previously been validated.

The present study aimed to evaluate the association between knee muscle strength and hop tests, referred to as muscle function tests, limb symmetry at the time of RTS to determine safe RTS, in athletes with ACL-R. Specifically, we aimed to differentiate athletes who have a safe RTS defined as no event of second ACL injury within 2 years after RTS from athletes who do not by:

1. Identifying the best-differentiating LSI cut-offs.
2. Evaluating predefined LSI cut-offs.

Based on the clinical assumption that being symmetrical, that is, LSI 100%, is protective of second ACL injury, meaning a safe RTS, we assessed the effect of the deviation from being symmetrical on the odds of a safe RTS by:

3. Assessing a 1% increment in LSI.
4. Identifying the best-differentiating LSI cut-offs in deviation from 100% symmetry.
5. Assessing a 1% increment in LSI deviation from symmetry.

METHODS

This observational cohort study used prospectively collected data from a rehabilitation registry, Project ACL. The study was reported using the REporting of studies Conducted using Observational Routinely collected Data guidelines.[20] Athletes and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Description of Project ACL

Project ACL is a rehabilitation registry based in Gothenburg, Sweden, which started in 2014. The registry is open for individuals with an ACL injury independent of treatment or level of activity. Participation in the Project ACL is voluntary, and the athletes can withdraw at any time. Athletes are provided with written information about participation and informed consent is obtained at registration. The baseline is set to the date of ACL injury, and if reconstructive surgery is performed, this becomes the new baseline. The athletes are regularly assessed with tests of strength and hop performance, referred to muscle function tests, and PROMs at 10 weeks, 4, 8, 12, 18, 24, 60 months and then every fifth year (figure 1). At each follow-up, athletes are asked to manually report the occurrence of a second ACL injury. Athletes report a second ACL injury by adding a new ACL injury to their profile, with date of injury and potential date of the second ACL reconstruction. The follow-ups then start over from baseline.[21]

Tests of muscle function

Muscle function tests include a test of muscle strength for knee extension and flexion, and three hop tests, that is, vertical hop, hop for distance and 30 s side hop. Strength assessments comprised unilateral concentric knee extension and flexion tests (0°–90°) performed in a seated isokinetic dynamometer, Biodex System 4 (Biodex Medical Systems, Shirley, New York, USA) with an angular speed of 90° per second. For all muscle function tests, the injured limb was tested first, followed by the non-injured limb. The athletes performed a standardised warm-up procedure of 10 min on a stationary bike and submaximal trials in the Biodex. Three to five single-repetition trials at maximal effort for knee extension and flexion are performed with 40 s in between each trial, where the peak torque in Newton metres (Nm) was recorded.[22]

Hop tests consisted of the vertical hop, hop for distance and the 30 s side hop test.[23] All hop tests were performed with the athletes holding their hands behind their backs. The vertical hop measure flight time, converted to centimetres (cm) using the Muscle Lab contact mat (Ergotest Technology, Oslo, Norway).

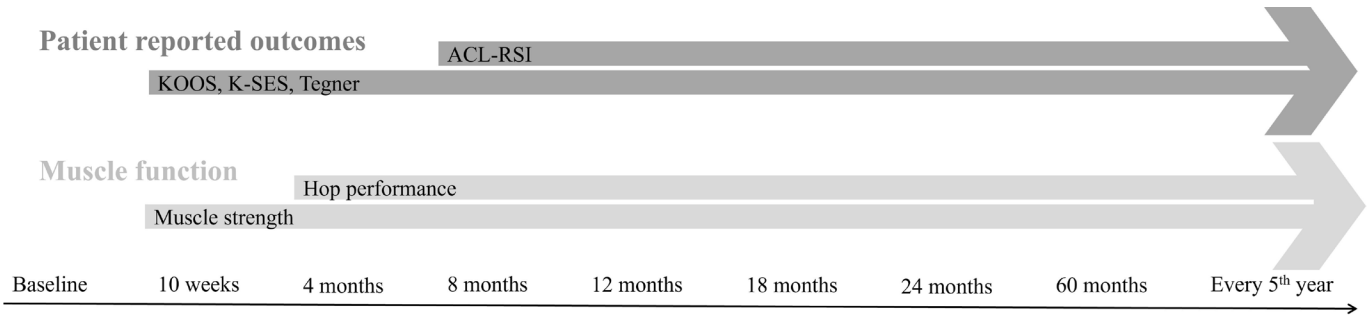

Figure 1 Graphical illustration of the follow-ups in Project ACL and what tests are performed at each follow-up. ACL-RSI, anterior cruciate ligament-return to sport after injury; KOOS, Knee injury and Osteoarthritis Outcomes Score; K-SES, Knee Self-Efficacy Scale, Tegner; Tegner activity scale.

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Athletes were instructed to perform three maximum hops for each limb. Hop for distance was measured in cm from toe at take-off to heel at landing. Athletes were required to perform a stable landing. A total of five maximum trials were allowed. The 30 s side hop test was performed past two lines, 40 cm apart. Athletes were given one trial on each leg, with 3 min rest between trials, and instructed to hop as many times as possible over the lines for 30 s. The best result for each hop test was recorded.

Patient-reported outcome measures

The Tegner Activity Scale[24] (Tegner) aims to provide a standardised method to determine the level of physical activity prior to injury and after injury that is documented on a numerical scale ranging from 1, walking on even ground, to 10, elite level sports. Tegner ≥6 represents the participation in pivoting sport and was used as a definition of athlete. The Tegner is a subjective scale where athletes rate their own activity level. The test–retest reliability of the Tegner has been reported as satisfactory, with an interclass correlation coefficient (ICC) of 0.81.[25 26] An ICC score >0.7 is regarded as satisfactory. In the present study, athletes were asked to rate their current Tegner levels at every follow-up, within 2 weeks from muscle function test scheduling (at 10 weeks, 4, 8, 12, 18 and 24 months). The definition of RTS for this study was the return to preinjury Tegner. The RTS time point was at or after the athlete had RTS. The Tegner does not record any information about exposure level to training or games or exposure time.

Athletes

Athletes eligible for inclusion were between 15 and 30 years old at index ACL-R, had a preinjury Tegner ≥6, that is, defined as an athlete in the present study, and had performed the Project ACL test battery at the time of RTS. Athletes were required to have 2 years of survival from RTS or had sustained a second ACL injury within 2 years of RTS being included. Athletes who had not performed an isokinetic strength test were excluded.

Definition of RTS

RTS was defined as return to the same Tegner level as preinjury. Athletes rated their Tegner and thereby their RTS within 2 weeks of every scheduled follow-up (at 10 weeks, 4, 8, 12, 18 and 24 months). The test in which athletes reported to have RTS is the test used for analysis.

Outcome measures

The outcome of interest in this study was a safe RTS addressed by using the LSI results from the test battery which include five tests of muscle function at the time of RTS. In order to achieve a safe RTS, new LSI cut-off values were calculated by the use of the receiver operating characteristic (ROC) and area under the curve (AUC) summarised with the Youden J and the accuracy of previous cut-off values was re-evaluated with the use of regression analyses presented as OR. Furthermore, for a safe RTS, new LSI cut-off values were defined in deviation from 100% symmetry use of the ROC and AUC summarised with the Youden J and evaluate the odds for a safe RTS in deviation from symmetry with the use of regression analyses presented as OR. The muscle function tests include two strength tests for the quadriceps and hamstrings muscle groups and 3 hop tests (vertical hop, hop for distance, and 30 s side hop). The test of when athletes first rate their RTS was used for analyses. We pursued our aim with five different approaches described as follows:

1. To identify LSI cut-off values for each of the muscle function tests that best differentiate athletes with a safe RTS from athletes who sustained a second ACL injury. In addition, we tested how well passing the test battery, with the newly determined cut-offs, was able to differentiate between athletes with a safe RTS and athletes who did not.
2. To test how well predefined LSI cut-offs, ranging between 80% and 90%, used across the test battery, were able to differentiate between athletes with a safe RTS and athletes who sustained a second ACL injury.

Based on the clinical assumption that symmetry, that is, LSI 100%, is protective of second ACL injury meaning in a safe RTS, we assessed how an increment in LSI affected the odds for safe RTS, the effect of the deviation from being symmetrical on the odds of a safe RTS. Deviation from symmetry (100%) was defined as a 1% increment, regardless of being positive or negative, on the odds of a safe RTS. For these analyses, having an LSI=100% was set as 0, and the deviation in per cent was used as a predictor, for example, an LSI of 90% would in this analysis be used as 10%, as 90% deviates 10% from 100%.

3. To test how well a 1% increment in LSI for all muscle function tests was able to differentiate between athletes with a safe RTS and those who did not.
4. To identify LSI cut-offs in deviation from symmetry (100%), for all muscle function tests, that best differentiate athletes with a safe RTS and those who did not.
5. To test how well a 1% increment in deviation, regardless of positive or negative from being symmetrical, for the muscle function tests was able to differentiate between athletes with a safe RTS and those who did not.

Equity, diversity and inclusion statement

This study included both women and men undergoing ACL-R. Our multidisciplinary research team involved women and men researchers from different medical areas (orthopaedics and physical therapy) and career levels (junior researchers to professors). Data were collected from the Project ACL and thereby, data collection did not differ between the included athletes. All athletes undergoing ACL-R, involving different ethnicities, cultures, socioeconomic levels and sexes are welcome to participate in the Project ACL.

Statistics

The SAS software V.9 (Copyright 2016 SAS Institute) was used for statistical analysis. Demographic data were presented as arithmetic mean values with SDs and medians with minimum and maximum values. A significance level of 0.05 was used.

To identify LSI cut-off values for each muscle function test, the ROC and AUC were used (aims 1 and 4). The ROC curve is a graphical method used to display the accuracy of a marker in differentiating between two groups, in this case, a safe RTS or sustaining a second ACL injury. The athlete was assessed as 'positive' if having the outcome of interest, that is, a safe RTS. The athlete was assessed as 'negative' if not having a safe RTS, that is, sustaining a second ACL injury.[27] The assessment of athletes being positive or negative was performed in the ROC plotting true positives (sensitivity) and the possibility of false positives (1−specificity). Results were summarised using two measures for the ROC, the AUC and the Youden J. The AUC is a measure of accuracy, that is, how well the model can distinguish between outcomes by using predefined cut-offs and newly defined best-differentiating cut-off values. The AUC reference values are 0.5=no differentiation value; 0.5–0.7=poor differentiation;

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Table 1 OR per 1% change of LSI, area under the ROC curve, predicting safe RTS within 2 years from RTS

Variable	Total n	Value	n descriptive subgroup	n (%) of safe RTS	OR (95%CI) safe RTS within 2 years from RTS	P value	AUC (95%CI)
LSI hamstrings	233	**66 to <96**	77	64 (83.1)	1.00 (0.97 to 1.04)	0.84	0.52 (0.42 to 0.62)
		96 to <104	78	66 (84.6)			
		104 to 135	78	66 (84.6)			
LSI quadriceps	233	**64 to <93**	77	68 (88.3)	0.98 (0.95 to 1.01)	0.18	0.59 (0.49 to 0.68)
		93 to <102	78	65 (83.3)			
		102 to 161	78	63 (80.8)			
LSI vertical hop	233	**51 to <87**	77	69 (89.6)	0.97 (0.95 to 1.00)	0.04	0.59 (0.50 to 0.69)
		87 to <98	78	64 (82.1)			
		98 to 149	78	63 (80.8)			
LSI hop for distance	233	**69 to <93**	77	66 (85.7)	0.96 (0.92 to 1.01)	0.11	0.59 (0.49 to 0.70)
		93 to <99	78	70 (89.7)			
		99 to 116	78	60 (76.9)			
LSI 30 s side hop	233	**38 to <93**	77	65 (84.4)	1.00 (0.97 to 1.02)	0.77	0.52 (0.43 to 0.62)
		93 to <102	79	68 (86.1)			
		102 to 139	77	63 (81.8)			
Achievement of specified LSI cut-off values							
All 90% LSI	233	**No**	158	136 (86.1)	0.65 (0.31 to 1.33)	0.24	0.55 (0.46 to 0.64)
		Yes	75	60 (80.0)			
All 85% LSI	233	**No**	106	96 (90.6)	0.39 (0.18 to 0.84)	0.016*	0.61 (0.53 to 0.69)
		Yes	127	100 (78.7)			
All 80% LSI	233	**No**	69	64 (92.8)	0.32 (0.12 to 0.87)	0.025*	0.60 (0.53 to 0.66)
		Yes	164	132 (80.5)			
All five LSI for insignificant newly determined cut-offs	233	**No**	202	172 (85.1)	0.60 (0.24 to 1.51)	0.28	0.53 (0.47 to 0.60)
		Yes	31	24 (77.4)			

All tests are performed with univariable logistic regression. P values, OR and area under ROC curve are based on original values and not on stratified groups.
Bold values present the range of achieved LSI values for each muscle function tests.
*p<0.05.
ACL, anterior cruciate ligament; AUC, area under the curve; LSI, Limb symmetry index; n, number of athletes; n (%) of safe RTS, number of athletes who can RTS safely; ROC, receiver operating characteristic; RTS, return to sport.

0.7–0.8=acceptable differentiation; 0.8–0.9=excellent differentiation and >0.9=outstanding differentiation.[28] The Youden J measures the effectiveness of the cut-off point summarised from the ROC curve.[27] The Youden J is derived from sensitivity plus specificity–1, where a result of 0 infers that the assessed cut-off is no better than random, while a result of 1 infers that the assessed cut-off perfectly separates athletes who have a safe RTS from those who sustain a second ACL injury.

A logistic regression analysis was performed with safe RTS as the dependent variable, for every specific test and the combination of passing all tests (independent variables) with the best differentiating, or predefined cut-offs (aims 2, 3 and 5). The logistic regression analysis was used to assess whether the cut-offs were associated with odds of safe RTS within two years after RTS. The logistic regression analysis was presented as ORs with 95% CIs. The p value and ORs were derived from a continuous analysis of a 1% increment in LSI or deviation from symmetry and are presented stratified by groups, as shown in tables 1 and 2 in this study.

RESULTS

Of the 2761 athletes available in Project ACL at the time of study, there were 261 athletes eligible for inclusion. After applying exclusion criteria, 233 athletes were included (figure 2).

Table 3 presents the demographics for the included athletes. There were 119 (51.1%) women and the mean time to RTS was 14.2±10.1 months. 37 (16%) athletes sustained a second ACL injury within 2 years after RTS, of which 24 (65%) were ipsilateral and 13 (35%) were contralateral. Most athletes participated in Tegner level 9 activities, such as football (36.9%). Athletes who sustained a second ACL injury, RTS at 12.0±4.1 months after ACL-R, while athletes with a safe RTS returned at 14.6±10.8 months. There were no statistically significant differences between groups in terms of demographics (table 3). The frequency of second ACL injuries in months after RTS is presented in online supplemental file 1.

Identify LSI cut-offs

Table 4 presents the newly determined LSI cut-off values that best differentiate athletes with a safe RTS from athletes who sustain a second ACL injury. The ROC curve for each specific test is presented in online supplemental file 2. The Youden J and AUC for the cut-offs were significant for the vertical hop (p=0.04), however, the Youden J showed no differentiation capacity (0.19). The LSI cut-offs for muscle strength, hop for distance and 30 s side hop were non-significant and below acceptable values. Despite that only one LSI cut-off was significant, and the other LSI cut-offs had non-significant differentiating values for safe RTS after ACL-R, they were tested across the test battery. 31 athletes passed all five best-differentiating cut-offs for LSI, of which 24 athletes had a safe RTS, and 75 athletes passed the ≥90% LSI cut-off, of which 60 athletes had a safe RTS (online supplemental file 3).

Predefined LSI cut-offs

Athletes who achieved 80% LSI or 85% LSI across the test battery had significantly lower odds of safe RTS (OR 0.32, 95% CI 0.12

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Table 2 OR per 1% change of LSI deviation from 100 and area under the ROC curve, predicting a safe RTS within two years from RTS

Variable	n	n missing	Value	n descriptive subgroup	n (%) of safe RTS	OR (95%CI) safe RTS within 2 years from RTS	P value	Area under ROC curve (95%CI)
LSI Hamstrings	233	0	**0.0 to <4.5**	78	66 (84.6)	0.98 (0.93 to 1.03)	0.38	0.50 (0.40 to 0.60)
			4.5 to <9.2	77	63 (81.8)			
			9.2 to 35.3	78	67 (85.9)			
LSI Quadriceps	233	0	**0.0 to <4.5**	77	63 (81.8)	1.04 (0.98 to 1.10)	0.16	0.59 (0.49 to 0.69)
			4.5 to <8.9	78	64 (82.1)			
			8.9 to 60.6	78	69 (88.5)			
LSI Vertical hop	233	0	**0.0 to <6.2**	77	63 (81.8)	1.01 (0.97 to 1.04)	0.70	0.55 (0.44 to 0.65)
			6.2 to <14.9	78	66 (84.6)			
			14.9 to 49.4	78	67 (85.9)			
LSI Hop for distance	233	0	**0.0 to <3.8**	77	61 (79.2)	1.04 (0.97 to 1.12)	0.22	0.56 (0.45 to 0.66)
			3.8 to <7.6	78	68 (87.2)			
			7.6 to 30.8	78	67 (85.9)			
LSI 30 s side hop	233	0	**0.0 to <4.2**	77	63 (81.8)	1.02 (0.98 to 1.05)	0.40	
			4.2 to <12.1	79	64 (81.0)			0.53 (0.43 to 0.62)
			12.1 to 61.9	77	69 (89.6)			
PASS all five muscle function tests, deviation from LSI 100	233	0	**No**	231	195 (84.4)	0.18 (0.01 to 3.02)	0.24	0.51 (0.48 to 0.54)
			Yes	2	1 (50)			

All tests are performed with univariable logistic regression. P values, OR and area under ROC curve are based on original values and not on stratified groups.
Bold values present the range of achieved deviation from 100% LSI values for each muscle function test.
ACL, anterior cruciate ligament; LSI, Limb symmetry index; n, number of athletes; n (%) of safe RTS, number of athletes who can RTS safely; PASS, passing newly determined cut-offs; ROC, receiver operating characteristic; RTS, return to sport.

to 0.87 p=0.025 and OR 0.39, 95% CI 0.18 to 0.84 p=0.016, respectively). The AUC values for these analyses showed poor predictive capacity (table 1). There was no significant association between passing 90% LSI and safe RTS.

Increment by 1% in LSI

There was a significant association for the 1% increment in LSI from the vertical hop (p=0.04), however, AUC values had poor differentiation capacity. There were no significant associations for the 1% increment in LSI from the individual muscle strength tests, hop for distance or 30 s side hop or across the test battery and safe RTS (table 1).

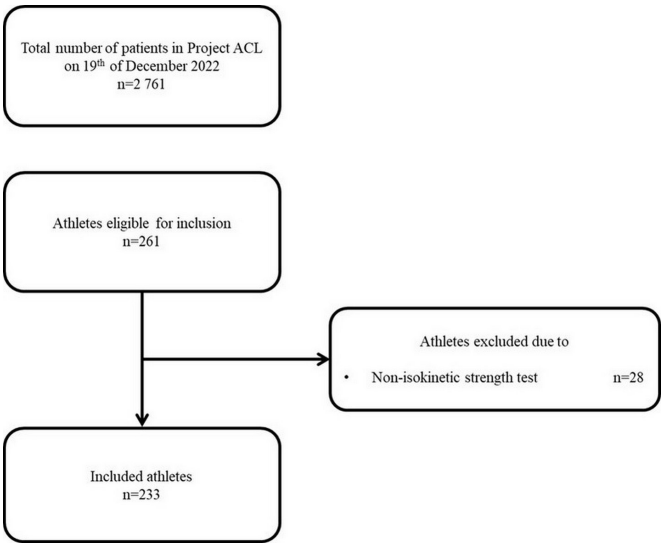

Figure 2 Flow chart on the study inclusion and exclusion.

LSI cut-off deviation from symmetry

None of the cut-offs for the analyses on LSI deviation from being symmetrical had significant differentiation values. Online supplemental file 4 presents the ROC curve for the cut-offs for each muscle function test. The Youden J and AUC did not reach acceptable values for the cut-offs for deviation from symmetry (table 5). Only two athletes passed all five muscle function tests using the best-differentiating cut-offs in deviation from symmetry (online supplemental file 5).

Increment by 1% deviation from symmetry

Table 2 presents ORs for every 1% change of LSI deviation from 100% symmetrical. No significant association was found for a safe RTS and the deviation from 100% LSI using neither the muscle function tests when analysed continuously, nor when passing the best-differentiating cut-off for all five tests.

DISCUSSION

We assessed commonly used and newly defined LSI cut-offs to determine a safe RTS for athletes after ACL-R within 2 years from RTS. The main findings of this study suggest that the use of LSI to interpret results from tests of strength and hop tests to determine safe RTS after ACL-R cannot differentiate athletes who have a safe RTS from athletes who do not, regardless of whether LSI was used as cut-offs, incremental or as deviation from symmetry. This implies that clinicians cannot rely on LSI alone for decision-making if athletes safely can RTS. We did, however, find lower odds for a safe RTS in athletes who passed LSI cut-offs for vertical hop, and for athletes who passed 80% LSI or 85% LSI on all muscle function tests. It is important to note that the potentially protective effect of not achieving 80% or 85% LSI had poor predictive capacity and considering the lack of other associations across the assessments on symmetrical

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Table 3 Athletes' demographics by safe RTS or not within 2 years from RTS

	Missing	Total population	Safe RTS	Second ACL injury	P value	Effect size
n	0	233	196	37		
Sex, n (%)						
Women	0	119 (51.1)	101 (51.5)	18 (48.6)		
Men	0	114 (48.9)	95 (48.5)	19 (51.4)	0.89	0.06
Age at time of reconstruction (years)	0	20.5 (4.0) 19.9 (15; 30)	20.5 (4.0) 19.9 (15; 29.8)	20.5 (3.9) 20.2 (15; 30)	0.93	0.01
Body height (cm)	0	175.6 (9.4) 175 (155; 200)	175.9 (9.6) 175 (155; 200)	174.1 (8.4) 174 (158; 193)	0.30	0.19
Body weight (kg)	1	72.9 (11.2) 71 (48; 108)	73.0 (11.2) 71 (48; 108)	72.6 (11.1) 72 (52; 95)	0.88	0.03
BMI (kg/m^2)	1	23.6 (2.5) 3.5 (17.3; 34.5)	23.5 (2.6) 23.4 (17.3; 34.5)	23.8 (2.3) 24.1 (19.8; 31)	0.43	0.13
Time to RTS (months)	0	14.2 (10.1) 11.9 (3.8; 120.4)	14.6 (10.8) 11.9 (3.8; 120.4)	12.0 (4.1) 11.9 (7.7; 24.3)	0.07	0.26
Tegner preinjury, n (%)						
6		9 (3.9)	8 (4.1)	1 (2.7)	0.38	
7		35 (15.0)	29 (14.8)	6 (16.2)		
8		73 (31.3)	65 (33.2)	8 (21.6)		
9		86 (36.9)	70 (35.7)	16 (43.2)		
10		30 (12.9)	24 (12.2)	6 (16.2)		
Type of graft, n (%)						
Hamstring		184 (79.0)	151 (77.0)	33 (89.2)	0.34	
Patellar		44 (18.9)	41 (20.9)	3 (8.1)		
Other		1 (0.4)	1 (0.5)	0		
Unknown		4 (1.7)	3 (1.5)	1 (2.7)		
Second ACL injury side, n (%)						
Ipsilateral		24 (64.9)	0	24 (64.9)		
Contralateral		13 (35.1)	0	13 (35.1)		

For categorical variables, n (%) is presented. For continuous variables mean (SD), median (min;max).
ACL, anterior cruciate ligament; BMI, body mass index; n, number of athletes; RTS, return to sport; Tegner, Tegner Activity Scale.

muscle function and safe RTS in this study, we consider these findings to have limited clinical relevance.

RTS testing builds on a theory to use a single snapshot of an athlete's strength and function to predict how well, or safely, they can participate in sports after ACL-R. Although naive, this has been used by rehabilitation specialists in the hope of better guiding decision-making on when an athlete can safely RTS, that is, with a minimal risk for second ACL injury.[5 29] The methodology of using RTS tests has become more common and has likely positively affected clinical practice to guide rehabilitation and provide a better understanding of knee function and readiness to RTS, both physical and psychological. At the same time, the single snapshot of an athlete's function is limited by not accounting for variability in performance during the testing session and over time, by the tests used in the test battery, and the transferability and validity between the specific tests to the 'chaos' of sports.[10] When discussing safe RTS, it is important to consider not only RTS test results but also the time to RTS. It is worth noting that patients in the present study RTS at 12–14.6 months, which is later than prior research that supported the use of LSI to help determine RTS, that is, ~7.5 months[12] and ~8 months.[7] Furthermore, athletes from a systematic review and meta-analysis,[30] who sustain a second ACL injury appear to RTS earlier compared with athletes who have a safe RTS. In our study, athletes with a safe RTS returned, on average, 2.6 months later than athletes who sustained a second ACL injury (p=0.07). This difference in time to RTS may suggest that the timeline of the biological healing process is clinically important, irrespective of RTS test outcomes.[31] However, a more suitable approach is to carefully consider the interplay between the time to RTS, involving biological healing and maturation, as well as physical and psychological preparation, and knee muscle function as indicated by the LSI. This balanced perspective should inform discussions regarding the RTS plan, facilitating informed decisions between athletes and clinicians.

Table 4 Cut-off values for LSI, Youden J, 1-specificity, sensitivity and model AUC for five tests of muscle function

Specific test	LSI cut-off	Youden J	Specificity	Sensitivity	Model AUC	P value
Hamstring	98.2	0.13	0.41	0.54	0.52	0.84
Quadriceps	97.6	0.24	0.49	0.73	0.59	0.18
Vertical hop	84.6	0.19	0.70	0.89	0.59	0.04
Hop for distance	97.1	0.21	0.41	0.62	0.59	0.11
30 s side hop	87.8	0.09	0.78	0.86	0.52	0.77

ACL, anterior cruciate ligament; AUC, area under the curve; LSI, Limb symmetry index.

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Table 5 Cut-offs, Youden J, sensitivity and specificity for deviation from LSI 100%

Deviation from symmetry	Cut-off	Youden J	1–specificity	Sensitivity	Model AUC	P value
Hamstring	1.8	0.14	0.84	0.97	0.50	0.38
Quadriceps	7.4	0.20	0.53	0.73	0.59	0.16
Vertical hop	17.3	0.15	0.72	0.86	0.55	0.70
Hop for distance	4.8	0.18	0.42	0.59	0.56	0.22
30 s side hop	12.2	0.15	0.66	0.81	0.53	0.40

AUC, area under the curve; LSI, Limb symmetry index.

LSI cut-off values

Despite the common use of limb symmetry to interpret results from muscle function tests to determine if athletes are ready to RTS, there is no current consensus on which cut-offs are the best for safe RTS.[7 10 12 15 32] Different cut-offs have been used throughout the ACL literature, where LSI ≥90% is probably the most common,[7] supported by recommendations from consensus statement[4 16–18] and expert opinions.[4 19] In healthy, non-ACL injured controls, the LSI for lower extremity strength[33] and hop tests[34] has been reported to be slightly higher and more symmetrical, compared with individuals after ACL-R. One reason for the limited consensus on which LSI cut-offs to determine safe RTS after ACL-R can be exemplified by this study. Our results imply that the use of LSI to interpret results from tests of muscle function, where we assessed commonly used cut-offs and the best-differentiating cut-offs in our data, provided no clinical use to identify which athletes safely can RTS after ACL-R. The mere use of LSI to interpret results from tests of muscle function appears to be simple to guide clinical practice, however, should future studies be assessed in combination with other factors, such as time to RTS and other qualitative and quantitative information on the lower limb function.[7 14]

Deviation from being symmetrical

In this study, we questioned the use of symmetry in muscle function to reflect recovery and its association with an athlete's ability to safely RTS. To do this, we assessed the assumption that being symmetrical, LSI=100% would imply that an athlete has a safe RTS, that is, with minimal risk for second ACL injury, and any deviation from being symmetrical would imply an increased risk of second ACL injury. We found that the Youden J and AUC did not reach acceptable values, suggesting that the deviation from symmetry cannot assist in determining a safe RTS after ACL-R.

Clinical implications: what we believe is the future for testing of athletes after ACL-R

Our findings, which challenge the use of symmetrical muscle function as a guide for RTS decision-making, warrant (1) confirmation in a replication trial on a different cohort and (2) a discussion on what and where we believe the field of RTS testing is heading. High-quality research has during a sufficient period of time implemented structured and repeated muscle function testing to move away from practice 'guessing' to 'measuring' in the hope of better guiding RTS decision-making.[35 36] We still believe that tests of muscle function have an important place in ACL rehabilitation as it enhances motivation in athletes and provides concreteness to the communication between healthcare professionals and athletes.

We also believe that it is a matter of what is measured. The fact that the demands placed on athletes vary depending on sport is one point that may need further consideration in terms of the tests used, that is, sports specificity and the athlete's anthropometrics.[37] Another point relates to the cognitive demands of the tests used to determine RTS after ACL-R, where we at some point may need to move away from the structured environment typically used, with low cognitive demands, to situations that mimic the cognitive stress which an athlete will be exposed to during sports, such as on-field rehabilitation.[38] There is no test that can account for all potential internal and external factors that constitute a safe RTS or not which will act on the athlete as time passes from the time of testing. We should remember that it is likely impossible to account for all injury risk factors, and successfully predict and prevent all prevalence of second ACL injuries. Therefore, we need to dig deeper into the results from this and other studies and ideally understand the isokinetic strength and hop testing performance for each individual athlete and what is required of them in their sport.

Limitations

We used the Tegner to determine RTS which is not without limitations. We used Tegner ≥6 as an inclusion criterion as this ensured that we only included athletes who before their ACL injury participated in pivoting sports. The use of return to preinjury Tegner, to determine RTS, provided reliable information that the athlete had returned to an activity that was considered as demanding as the athlete's preinjury exposure. The Tegner does, however, not give details on the exact type of activity, intensity of training or the time of athletic exposure. Whether or not athletes had returned to unrestricted performance or if RTS reflected restricted participation is therefore unknown. Furthermore, the rating of the Tegner was made at the time of scheduled follow-ups (10 weeks, 4, 8, 12, 18 and 24 months) while athletes could have RTS between follow-ups. This means that athletes may have RTS at 7 months but did not have their Tegner rating updated until the 8-month follow-up. The possible gap between RTS and follow-up may also infer there is a difference in muscle function at time of the actual RTS and what is displayed at the first rating of RTS, which is a major limitation for this study. The RTS time point was at or after the athlete had RTS. In addition, we did not take concomitant injuries into account as these are not registered in Project ACLs database. Concomitant injuries such as cartilage and meniscal injuries could affect a patient's possibility to achieve a symmetrical LSI and thereby the results of the present study. In addition, previous studies have shown that the uninjured limb may decline in muscle strength and hop performance[8 39] but also increase in muscle strength after ACL-R.[40] Absolute values of the injured or the non-injured limb were not considered in this study. The LSI represents the patient's symmetry in function, regardless of the test used to assess strength, therefore, we have not considered whether results were affected by changes in absolute values. Furthermore, we calculated the

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LSI from the peak torque of the quadriceps and hamstrings strength assessed isokinetically and concentric in a seated position, which may not provide the complete picture of muscle strength. Ebert *et al*[41] reported LSI differences through the torque curve despite having LSI ≥90% assessed isokinetic concentric in athletes after ACL-R, which led authors to conclude that analyses of knee extension and flexion torques should be made throughout full range of motion. Findings in Ebert *et al*[41] study has further been supported in another study by Tengman *et al*.[42] Hence, the reliability and external validity of using peak torque from isokinetic strength testing can be discussed in a more thorough investigation of muscle strength compromising both full range of motion, eccentric and rate of force development motions.

CONCLUSION

The use of LSI to interpret results from a test of muscle function to determine safe RTS, that is, RTS without sustaining a second ACL injury, after ACL-R cannot differentiate between athletes who had a safe RTS from athletes who did not, regardless of whether LSI was used as cut-offs, incremental or as deviation from symmetry, within 2 years from RTS. Thus, it is of clinical importance that clinicians do not solely rely on the LSI to clear athletes for RTS.

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